

Total No. of Questions : 4]

SEAT No. :

PE48

[6579]-350

[Total No. of Pages : 1

T.E. (Electronics and Telecommunication) (Insem)

ELECTROMAGNETIC FIELD THEORY

(2019 Pattern) (Semester - I) (304182)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4.
- 2) Neat diagram must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Use of electronic packet calculator and smith chart is allowed.
- 5) Assume suitable data, if necessary.

- Q1) a) State and Prove Gauss's Law. [6]
b) The electric potential near the origin of a system of co-ordinates is $V = 5x^2 + 8y^2 + 10z^2$. Find the electric field at a point A (1,2,3). [6]
c) Define Electric Field intensity and derive its expression for point charge. [3]

OR

- Q2) a) Two point charges $-4 \mu\text{C}$ and $5 \mu\text{C}$ are located at (2,-1,3) and (0,4,-2) respectively. Find the potential at (1,0,1) assuming zero potential at infinity. [6]
b) Derive an expression for the potential gradient $E = -\nabla V$. [5]
c) Explain the physical significance of gradient and divergence. [4]

- Q3) a) State Ampere's circuital law. Find magnetic field intensity H due to an infinitely long straight current carrying conductor using Ampere's law. [7]
b) The point Q=18 nC is has velocity of $5 \times 10^6 \text{ m/s}$ in the direction $a_v = 0.60 a_x + 0.75 a_y + 0.30 a_z$ calculate the magnitude of force exerted on the charge by the field, [6]

- i) Electric field intensity $E = -3 a_x + 4 a_y + 6 a_z \text{ kV/m}$
- ii) Magnetic flux density $B = -3 a_x + 4 a_y + 6 a_z \text{ mT}$

- c) Define and explain Biot-Savart's law. [2]

OR

- Q4) a) Find the vector current density in rectangular coordinate system at A (2,3,4) if Magnetic field intensity $H = x^2 z a_x - y^2 x a_y + z^2 y a_z \text{ A/m}$. [6]
b) Find magnetic field intensity H due to a circular current carrying loop on the axis of a loop using Ampere's law. [7]
c) State and explain Lorenz force equation. [2]